

EE201: ELECTRICAL CIRCUITS AND NETWORK
CREDITS = 6 (L=4, T=0, P=2)

Course objective:

The subject aims to provide the student with:

C1: Understanding of concepts and principles of passive circuit analysis and synthesis

C2: Ability to solve complex circuits using different theorems and methods.

C3: Advanced understanding of electrical networks which will be useful for advance Subjects.

Teaching and assessment scheme:

Teaching Scheme			Credit	Marks Distribution				Total Marks
L	T	P	C	Theory Marks		Practical Marks		
4	0	2	6	ESE	CE	ESE	CE	
				70	30	30	20	150

Course contents:

Unit No.	Topics	Teaching Hours
1	<u>Circuit Variables and Circuit Elements and Sources:</u> E.M.F, Potential and Potential Difference, Current and Current Density, Ideal and Practical Voltage and Current Sources, Conversion from one source into other, Internal Impedance of voltage and current source relative to load, Independent and Dependent Electrical Sources, Power and Energy Relations for Two-terminal Elements, Dot Convention, Duality.	05
2	<u>Nodal Analysis and Mesh Analysis of Resistive Circuits:</u> Nodal Analysis of Circuits Containing Resistors and Independent and Dependent Sources, Source Transformation Theorem for circuits with independent sources, Source Transformation Theorem for circuits with Dependent sources , Nodal Analysis of Circuits Containing Dependent Sources, Mesh Analysis of Circuits with Resistors and Independent Voltage Sources, Mesh Analysis of Circuits with Independent Sources, Mesh Analysis of Circuits Containing Dependent Sources.	06

3	<u>Circuit Theorems and Their Application in Electric Networks:</u>	06
	Linearity of a Circuit and Superposition Theorem, Substitution Theorem, Compensation Theorem, Thevenin's Theorem and Norton's Theorem, Reciprocity Theorem, Maximum Power Transfer Theorem, Millman's theorem.	
4	<u>Initial Conditions:</u>	04
	Initial conditions in elements, procedure for evaluating initial conditions, Solution of circuit equations by using Initial Conditions.	
5	<u>Time domain response of First order RL and RC circuits:</u>	05
	Mathematical preliminaries, Source free response, DC response of first order Circuits, Superposition and linearity, Response Classifications.	
6	<u>Time domain response of Second order linear circuits:</u>	04
	Discharging of a Capacitor through an inductor, Source free second order linear networks, and second order linear networks with constant inputs.	
7	<u>Laplace Transform Analysis and Circuit Applications:</u>	05
	Notions of Impedance and Admittance, Manipulation of Impedance and Admittance, Notions of Transfer Function- Equivalent circuits for inductors and capacitors, Nodal and Loop analysis in the s-domain, Switching in RLC circuits, Switched capacitor circuits and conservation of charge.	
8	<u>Laplace Transform Analysis and Transfer Function Applications:</u>	04
	Poles, Zeros and the s-plane, Classification of Responses, Computation of sinusoidal steady state response for stable networks and systems.	
9	<u>Two –Port Networks :</u>	06
	One port networks, Two port admittance Parameters (y parameters), Admittance parameters analysis of terminated two Port networks, Two port impedance Parameters (z-parameters), Hybrid parameters (h Para), Inverse Hybrid Parameters (g-para), Transmission parameters (ABCD parameters), Various Combinations of Two-Port network.	
10	<u>Introduction to Network Topology:</u>	06
	Linear Oriented Graphs (Connected Graph, Sub graphs and Some Special Sub graphs), The Incidence Matrix of a Linear Oriented Graph, Kirchhoff's Laws in Incidence Matrix Formulation– The Circuit (Loop) Matrix of a Linear Oriented Graph- Kirchhoff's Laws in Fundamental Circuit (Loop) Matrix	

Formulation, The Cut-set Matrix of a Linear Oriented Graph , The Fundamental Cut-set Matrix , Relationship among various Matrices.

11 **Introduction to circuit simulation using various software's.** 04

TOTAL 55

List of references:

1. Franklin S. KUO, "*Network Analysis & Synthesis*", Wiley Publication
2. M.E Van Valkenburg, "*Network Analysis*", PHI Publication
3. K. S. Suresh Kumar, "*Electric Circuits and Networks*" Pearson Education
4. U.A. Patel, "*Circuits and Networks*", Mahajan Publications

Course outcomes (COs):

After study this subject students will be able to ...

- CO1. Design basic hardware circuits
- CO2. Apply various circuit laws and methods like Ohm's Law, KVL, KCL, dot convention technique node and mesh circuit analysis
- CO3. Employ various network theorems such as Superposition, Thevenin, Norton, Reciprocity, Maximum Power Transfer, Millman's Theorem, etc to solve a complicated network
- CO4. Analyze behavior of passive circuits such as RC, RL and RLC taking into account initial conditions
- CO5. Apply Laplace Transform for circuit analysis and to obtain transfer function of a network.
- CO6. Calculate two port parameters such as y, z, h, ABCD, etc. for the given two port network and to write metrics for network topologies